THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 3: **Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Sodium is stored in kerosene oil. Reason (R): Sodium reacts vigorously with oxygen and moisture of the air. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Metal X reacts with dilute HCl giving H_2 but does not react with cold or hot water. Metal Y reacts with cold water. Which is more reactive? 1

(a) Y **(b)** X **(c)** both equally **(d)** cannot be said

3. A student heats a green carbonate ore of copper. The black residue and gas are: 1

(a) Cu and CO_2 **(b)** Cu_2O and O_2
(c) CuO and CO_2 **(d)** CuO and CO

4. Assertion (A): Aluminium is a highly reactive metal, yet it is used to make cooking utensils. Reason (R): Aluminium forms a thin, impervious oxide layer on its surface. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Four metals P, Q, R, S: P displaces Q from its salt; Q displaces R; R displaces S. The least reactive is: 1

(a) Q **(b)** R **(c)** S **(d)** P

6. Green copper carbonate on heating gives black solid X. X, when heated in hydrogen, gives a brown metal. The metal is: 1

(a) iron **(b)** lead **(c)** zinc **(d)** copper

7. Assertion (A): Ionic compounds are brittle. Reason (R): In ionic crystals, like ions can come near each other on a slight shift, causing repulsion. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A piece of Zn is placed in solutions of $FeSO_4$, $CuSO_4$ and $MgSO_4$. Reaction will occur in: 1

(a) $FeSO_4$ and $CuSO_4$ **(b)** all three
(c) $MgSO_4$ only **(d)** $CuSO_4$ only

9. **Assertion (A):** Sodium is stored under kerosene oil. 1
Reason (R): Sodium reacts vigorously with oxygen and moisture of the air and may catch fire.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Aluminium oxide is an amphoteric oxide. 1
Reason (R): Aluminium is widely used for making cooking utensils.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A metal X reacts with dilute HCl and also with NaOH producing hydrogen. Identify the nature of X's oxide with an example metal. 2

12. Why are metals in the middle of the reactivity series usually obtained from their oxides rather than sulphides or carbonates? 2

13. Why does an aluminium vessel not corrode easily though aluminium is above zinc in the reactivity series? 2

14. Why is the reaction of sodium with water not carried out in the lab with large pieces? Give the equation. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. In the reaction $\mathrm{ZnO + C \longrightarrow Zn + CO}$ identify the oxidising agent, reducing agent and the type of reaction. Why is it used for zinc? 3

16. Why does iron rust faster near the sea? Explain the role of salt in rusting. 3

17. Explain why ionic compounds like NaCl conduct electricity in aqueous solution while sugar solution does not. 3

OR

A student placed metals A, B, C, D in $\mathrm{FeSO_4}$, $\mathrm{CuSO_4}$ solutions. A and B displaced Fe and Cu; C displaced Cu only; D displaced none. Arrange in order of reactivity. 3

18. A black powder X on heating with hydrogen gives a brown metal Y. Y on heating in air gives X again. Identify and write equations. 3

OR

Explain why some metal oxides dissolve in water but copper oxide does not. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In an experiment, equal-sized pieces of magnesium, zinc, iron and copper were dropped into separate test tubes containing dilute hydrochloric acid. Bubbles of a gas appeared vigorously on magnesium, slowly on zinc, very slowly on iron and no bubbles appeared on copper. 5
(i) Which metal does not react with dilute HCl?
(ii) Name the gas evolved and give its test.
(iii) Write the equation for the reaction of zinc with dilute HCl.
(iv) Arrange the four metals in decreasing order of reactivity and state the conclusion drawn from the rate of bubbling.

20. Explain the extraction of aluminium from bauxite: steps, equations at electrodes, and why cryolite is added. 5

OR

A black coated copper wire is heated with hydrogen. Explain what is observed, write equations, and name the oxidising and reducing agents. 5

END OF QUESTION PAPER Total: 40 Marks